Beyond the Baseline: Stress-Testing AnomalyGFM

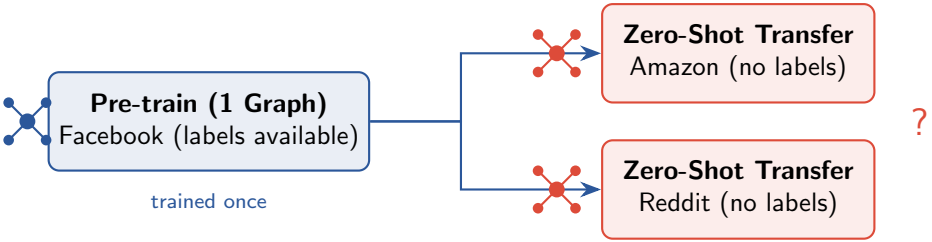
A Scientific Autopsy of Graph Foundation Models in Transfer Learning

Group 13

November 12, 2026

The Promise of AnomalyGFM

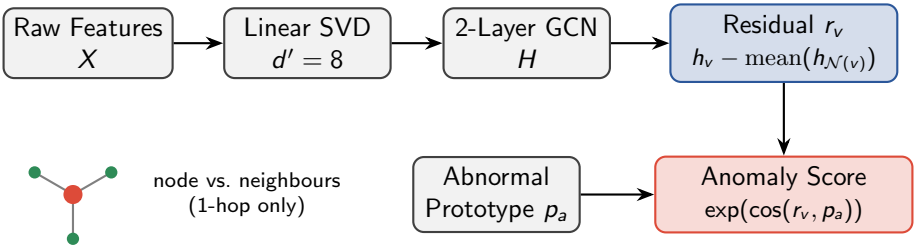
The Goal: Train a Graph Foundation Model once, detect anomalies zero-shot anywhere.



The Reality: Generic models fail here because anomalies are rare, irregular, and highly heterogeneous across domains.

Under the Hood: How it Actually Works

Core assumption: Normal nodes look like their 1-hop neighbors. Anomalies do not.



The Catch: One *single* prototype, a *1-hop* view, and an adjacency matrix assumed *100% honest*.

The Fragility of the Foundation

We audited the baseline architecture and found four blind spots:

Blind Trust in Adjacency
No camouflage assumed: fraudsters wired to normal nodes shrink their residual to zero.

Prototype Collapse
Diverse fraud types (structural vs. attribute) squeezed into one prototype blur the signal.

Rigid Receptive Field
The strictly 1-hop view misses deep laundering rings.

The Label Paradox
Pre-training needs labeled anomalies, contradicting the zero-shot premise.

Our Diagnostic Arsenal

We built four interventions to patch these structural flaws:

1

Defeating Camouflage
Edge Purification (C3): differentiable edge dropping prunes adversarial edges *before* message passing.

2

Capturing Diversity
Mixture-of-Experts (C1): $K=4$ experts replace the single prototype, with a dynamic routing gate.

3

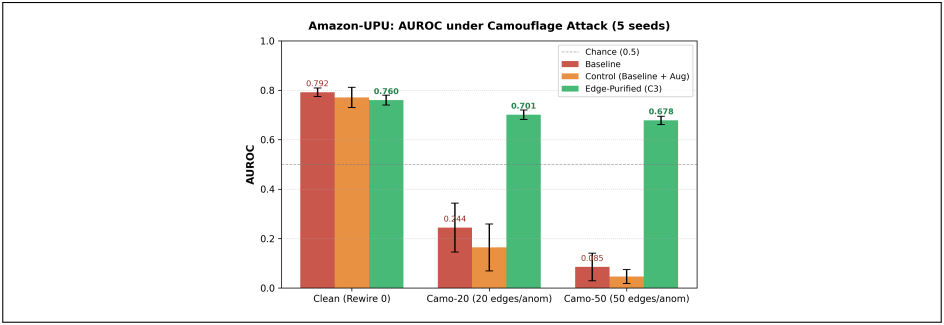
Label-Free Pre-training
Contrastive SSL (C3): synthetic structural/contextual corruptions learn prototypes with **zero labels**.

4

Target Adaptation
Unsupervised TTA (C2): entropy minimization on pseudo-labels, plus efficient **Graph-LoRA**.

Victory 1: Defeating Adversarial Camouflage

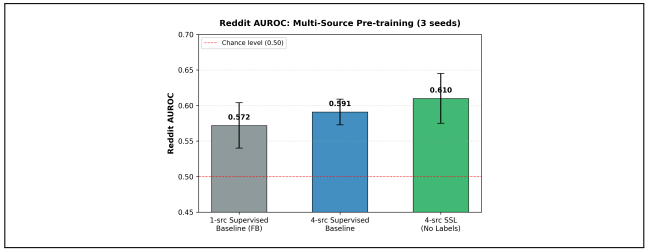
When anomalies hide behind normal nodes, the baseline is destroyed. Our Edge Purifier survives.



Amazon-UPU. Dashed line = chance (0.5). Under a 50-edge attack the baseline ranking inverts (0.085); the purifier drops 99.5% of injected edges.

Victory 2: Eradicating the Label Dependency

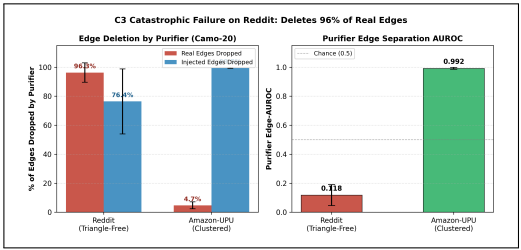
Anomaly structure can be learned completely in the dark (zero labels).



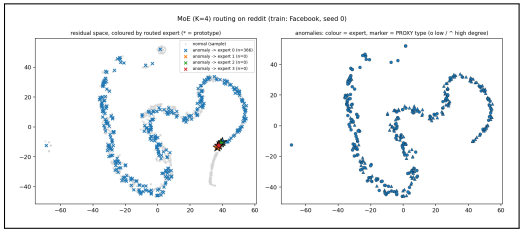
- **0.62 AUROC** on Amazon from synthetic rewiring and feature swaps alone.
- Best overall method on Reddit with multi-source pooling.

The Structural Transfer Wall

Why didn't advanced architectures work everywhere? **Local topology doesn't always transfer.**



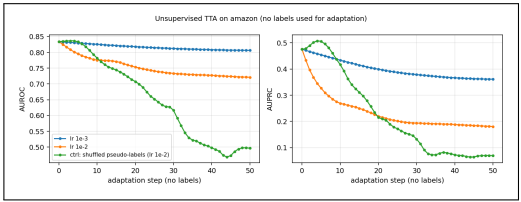
Purifier failure: Reddit is triangle-free. The Facebook-trained purifier learned "no shared neighbors = fake" and deleted 96% of real edges.



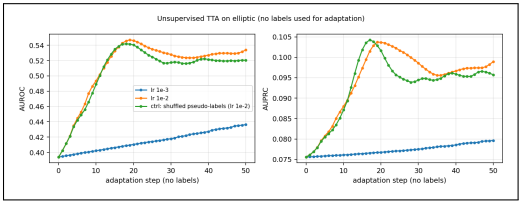
MoE collapse: Expert 0 takes 100% of Reddit anomalies. The router cannot generalize from only ≈ 7 labeled anomalies.

The Adaptation Paradox

Unsupervised Test-Time Adaptation faces a mathematical catch-22.



Good graphs (Amazon): labels are already 75% accurate, so sharpening an already-good distribution only adds noise.



Bad graphs (Elliptic): zero-shot ranking is inverted; pseudo-labels are worse than chance. The model learns from pure garbage.

What We Can and Cannot Claim

We CAN Claim

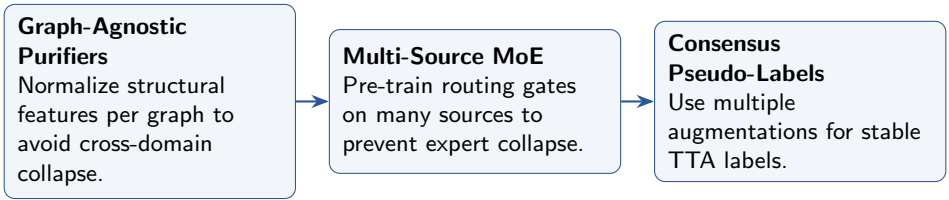
- **Camouflage breaks the baseline** (ranking inverts); edge purification fixes it on clustered graphs.
- **Label-free learning works** via synthetic corruptions.
- **Graph-LoRA is efficient:** matches full fine-tuning with 2.9% of the parameters.
- An adaptive PPR gate learns a per-node hop choice.

We CANNOT Claim

- **An average accuracy gain:** no extension beat the baseline on all targets.
- That MoE experts are real anomaly "types" (no type labels exist).
- That the purifier works on triangle-free graphs (Reddit) or under heavy random noise.
- Generality beyond the tested targets.

Conclusion & The Road Ahead

Verdict: The bottleneck for Graph Foundation Models is not model capacity, but the limit of single-graph transfer learning and the assumption of structural homogeneity.



Fully reproducible codebase: `github.com/SumitKumar-17/AnomalyGFM`
(Branch: team-contributions, Folder: Anomaly2/)